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Evidence-Based Global Public Health

Subjects, Profits, Erasures

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It's true that JAMA seldom publishes qualitative studies. A qualitative study can say that an exposure-outcome relationship occurs at least occasionally, and can describe some of the circumstances, but it cannot determine incidence, prevalence, independent risk factors, effective prevention or treatment, etc. At JAMA we are much more interested in studies at the other end of the research pipeline, meaning studies that show how to prevent or treat a known health problem.

—*Editor of the Journal of the American Medical Association, in response to a query about submitting a qualitative, ethnographic article on violence and human rights in reproductive health, 2010*

The topic of this chapter is the impact of “evidence-based medicine” (EBM) on global public health.¹ An epistemic transformation in the field of global health is underway, and I will argue that **the impact of evidence-based medicine has been twofold: (1) the creation of an experimental metric as a means of providing health care** (following Petryna 2009);² and (2)

a shift in the priorities of caregiving practices in public health such that “people [no longer] come first.” The production of experimental research populations in and through EBM helps constitute larger fiscal transformations in how we do global health. Notably, EBM has created a platform for the buying and selling of truth and reliability, abstracting clinical caregiving from the social relationships on which they depend.

The Rise of Evidence-Based Medicine

Most contemporary health scientists would agree that evidence-based medicine, as currently known, emerged in the 1990s (Timmermans and Berg 2003; Timmermans and Mauck 2005; Sackett et al. 2000; Evidence-Based Medicine Working Group 1992), although some have noted that the foundations of EBM may be located in the mid-nineteenth-century (Sackett et al. 1996) rise of scientific epistemology based on inductive empirical reasoning (see Gordon 1988) and the creation and use of an “experimental field” (see Shapin and Schaffer 1986). Modern EBM’s technical *raison d’être* has primarily been to address the persistent problem of nonstandard variation in clinical practice. The assumption in EBM is that a “conscientious, explicit, and judicious use of current *best evidence* in making decisions about the care of individual patients” will eliminate this problem and improve health care delivery and outcomes (Timmermans and Mauck 2005).³ In sum, the goal has been to create a stronger “scientific” foundation for clinical work by focusing on new kinds and meanings of “evidence.” I note at the outset that many have pointed to other compelling forces behind the birth of EBM, such as the rise of mandated cost control and resource limitations under “managed care” systems in industrialized nations. The perceived benefits of evidence-based medicine are significant and worthwhile, despite skepticism among some who doubt that the ideals of EBM will be achieved.⁴ Here I will focus on the rise of EBM in terms of its outcomes and implications beyond these debates.

Of importance in the shift to an era of “evidence-based medicine” is that while evidence can take many forms, in EBM, the statistical, experimental, and epidemiological models of evidence are taken as the *gold standard*. In fact, evidence in EBM is codified: “Level 1” evidence in EBM (on a scale of 5) is derived from the properly designed randomized controlled

trial (RCT). “Level 5” evidence, the least useful, derives from “opinions of respected authorities, based on clinical experience, descriptive studies, or reports of expert committees.”⁵ Between these extremes are the kinds of evidence derived from what EBM writers consider quasi-well-designed studies (controlled, cohort, case-control, multiple time series, etc.), but which are missing some critical component of the gold standard. In this approach to health care, the type of evidence that counts the “least,” if at all, derives from what gets called “anecdotal” information and is relegated to a level below 5. Like the untouchables of India’s caste system, “anecdotal” evidence is seen to lie beyond the pale of quality. “Anecdotal” refers to information based on the observations of nonclinical experts, unsystematic interviews, and eyewitness testimonials (Brownson et al. 2003).⁶ This kind of evidence is interesting and empirical, but not useful in the RCT. In EBM, studies that foreground the individual speaking subject as the primary source of truth have virtually no purchase, nor do those additional truths garnered from the families, communities, or relationships that help form that speech. Certain kinds of evidence are essentially irreconcilable with EBM’s definition of reliable evidence, except insofar as such words can be made to fit in a quantitative way into the RCT design, or otherwise to corroborate RCT findings with a bit of ethnographic “color.”

That the use of EBM is on the rise is clear. The epigraph at the start of this paper was written by the editor of the *Journal of the American Medical Association* in response to a colleague whose ethnographic research on the politics of sex selection and gender discrimination in assisted reproduction in the United States was considered inappropriate (or at best a long shot) for the journal. The editor states explicitly that “a qualitative study can offer evidence that something occasionally occurs,” but it cannot establish *reliable* evidence of the sort that really counts when it comes to effective prevention or treatment.⁷ Years of achievements of health policy, not to mention feminism, are summarily cast upon the ash heap of irrelevance here. This is not to say that I disagree entirely with the editor, for it is evident that some things indeed cannot be known except through techniques akin to RCT research, and journals like *JAMA* have *always* prioritized quantitative over qualitative studies. What is new today, however, and hinted at in the editor’s response, is that for evidence to be *reliable* today, it must be pushed through the engines of scientific scrutiny in

new ways. Specifically, for evidence to say anything valid about “how to prevent or treat a known health problem” it must speak the language of statistics and epidemiology, or, to be even more specific, it should use the methods of the randomized controlled trial. In this ordering of priorities, one finds a simultaneous discrediting of other forms of knowledge and evidence and other ways of conveying truth that have historically been invested with value in both deciphering ill health and evaluating the clinical outcomes of health interventions, especially in the global health context.

Viewed from the perspective of EBM advocates like *JAMA*’s editor, only randomized controlled studies can show that an intervention *really* works. RCT study designs are the most reliable way to be certain that what one is seeing is more than either a matter of chance or subjective perception and more than the occasional experience of a small and statistically insignificant group of individuals. RCT research knows its findings are reliable. Why? Because RCTs tie efficacy to statistical truth. That is, results of RCT studies are considered objective (they appear as unbiased, numerical facts) and they are considered valid (they use statistical configurations to present their truths). To contest these truths, one would have to doubt the reliability of both statistics and the objective realities they reveal. Although social science critics have contested this view, noting that the outcomes of such studies are in some sense prefigured through the anticipatory logic of the statistical method (that is, their outcomes are produced by the very structure of the research design as opposed to being discovered by it),⁸ EBM advocates take the statistical method as the most unbiased technique for evaluating health interventions.

To date, fundamental challenges that compel the shift to evidence-based methods in health care have been largely articulated around concerns over the validity and generalizability of empirical claims about outcomes. RCT models of research provide a convenient and reliable way of encoding and specifying a language for overcoming these challenges and achieving the goals of Best Medical Practices. As a result, it becomes increasingly the case that medical interventions that do not lend themselves to evaluations using RCT designs must be seen, from the EBM perspective, as being of *poor quality* and, by default, unreliable, and therefore of little use in determining policy, practice guidelines, or fiscal support.

I suggest that EBM approaches have become not just routine but rather tyrannical in the world of clinical health care in the industrialized nations. Not surprisingly, they are aggressively promoted as the new gold standard, not just in clinical medicine, but in territories that spread far beyond the US bedside to a wide variety of disciplinary and interventionary fields, including public and global health. The review panels that vet research grants and make funding decisions in global health (from large governments, bilaterals, and multilaterals, as well as foundations) have trended toward favoring projects that employ RCT in the belief that results from these projects will be more reliable because they can be more easily evaluated. While it is not entirely surprising that this is the case for research, it is surprising that health intervention projects are being subjected to similar demands to use RCT-style designs. In the United States, EBM is used to determine the best clinical practices and the optimal insurance policies. In global health, we are seeing similar trends with equally strong impact at intervention and policy levels. Even traditional humanitarian interventions that involve no research component are increasingly called to task over the question of whether their impact and effectiveness can be assessed using evidence-based standards of evaluation. New regimes of accountability using EBM standards have emerged as a kind of “audit” necessity in organizations far and wide, from the Gates Foundation and Save the Children, to the NIH and the WHO. The results of this shift are worth exploring.

In what follows, I recall the effort begun years ago by the World Bank anthropologist Michael Cernea in his volume on international development, *Putting People First*, as I attempt to answer the question posed by our conference organizers: What kind of medicine and what kind of research are called for if we wish to ensure that people come first? Subsidiary questions emerge from this as well: What kind of subjectivity must be produced in order to do EBM in global health? What kinds of economic engines are fed by this shift and who profits from them? What kinds of evidence must be disregarded in order for RCT-style evidence to matter more in the shift to evidence-based medicine in global public health? In the shift to valorizing one kind of evidence over other kinds of evidence, the unintended outcomes that arrive in the wake of this new regime of truth making in global public health need to be accounted for. This essay is only a first step in that effort.

Evidence-Based Medicine in Global Public Health: Producing Subjects

The contours of the transition to EBM in global public health settings are instructive and suggest an impact that is both epistemological and practical. First, EBM arrives in international health as part of the uptake of EBM by public health (in EBPH [evidence-based public health]) and alongside a shift in the field away from traditional “international health” toward what is called “global public health” or “global health science” or simply “global health” (Adams et al. 2008).

At my own university, a new multicampus Global Health Sciences program has been recently launched. Among its many rationales is the notion that clinicians who have long worked in the fields of health development will benefit from interchanges with bench scientists, pharmaceutical researchers, and industrialized nation public health experts as they grapple collectively with some of the most intractable problems of poor health on a global scale. Biochemists and molecular biologists undertaking research on infectious diseases find it increasingly necessary to work in globally remote sites in order to fully understand the etiologies and pathologies of viral, retroviral, and bacteriological pathogens and treatments that have a large and productive life outside of their laboratories. Infectious diseases travel rapidly around the globe, and researchers are finding that treatment programs designed for one location are useless in others. Such travels bring these bench researchers into contact with international health experts who have spent decades trying to eradicate and prevent transmission of these diseases in places where they have become endemic.

Pharmaceutical researchers are also finding themselves increasingly involved in international health work. Often, through the collaborative infrastructure of public-private partnerships, these researchers struggle to locate drug-naïve populations for clinical trials and to explore the dynamics of drug therapies among populations that are often in dire need of these drugs. At the receiving end, public and international health experts in underdeveloped and resource-poor countries find themselves having to negotiate with pharmaceutical research projects run by pharmaceutical contract research organizations (CROs) and with federally funded pharmaceutical research teams from wealthy nations (Petryna 2009; Biehl 2008). Together, they must navigate through, or around, existing

government-run public health infrastructures and policies in countries where research subjects are in plentiful supply but where thorny concerns about ethics, inclusion, and risk are still being worked out (Petryna et al. 2006; Petryna 2009).

Bringing models of traditional bench and pharmaceutical sciences research to international health is also in part a result of the perception that more scientific rigor has been needed in the world of international health, just as it was and is needed in clinical, bedside medicine at home. The meeting of bench and pharmaceutical scientists with international health experts has produced many fruitful and productive collaborations, but among the most significant is a “scientization” of international public health. That is, Global Health, as a new field emerging from the foundation of International Health Development, carries with it the notion that EBM-driven programs will satisfy the need for more scientific rigor in their international health development efforts. The use of evidence-based medicine will make global health look and feel more scientific because it will *be* more scientific in its practices. Already, most programs in global health have begun to train a cadre of intervention-oriented researchers who can think through international health by way of RCT-based languages and skills.

At the same time, the marriage of bench science and pharmaceutical researchers with traditional international health programs, and the scaling-up of use of RCT-model research in this effort, has also resulted in a shift away from certain forms of taken-for-granted “caregiving.” In place of the intervention programs that aimed at eradicating and treating disease by way of a variety of efforts aimed at maximizing impact for the largest numbers of people, a more ambitious mandate now circulates in global health, calling for “intervention-as-research” that aims to answer not the simple question “What works to improve health for the most people possible?” but rather “How can we be sure that our outcomes are not an artifact of chance or personal bias in perception?” The simple counting of those who are helped or made healthy is today insufficient, because now most, if not all, international health care interventions are asked to be attentive to a larger set of methodologies that allow for greater precision in the assessment of outcomes. Measurements modeled after the RCT enable researchers to determine the *statistical* significance of outcomes in ways that are believed to show evidence of impact that is uncontestable. Along

the way to getting this kind of evidence, however, some things might be compromised in the very prioritizing of what interventions can or should be done.

From the perspective of those who like to use RCT models, these methods are valuable because they offer a comparative platform that produces reliability based on statistical constants. The RCT proves efficacy in a number of quite specific ways—through the use of controls, randomization, generalizability, power calculations, and, above all, the use of only quantitative forms of data. RCTs define the variables that matter in terms of what can be counted, disregarding other kinds of information that might be attached to the intervention or outcome as irrelevant to the study. Using models that can be replicated in other geographic areas, other villages, other countries, EBM public health researchers believe that they will be able to demonstrate the efficacy of some interventions over others with a higher level of accuracy than in interventions where efficacy was claimed but results could not be replicated over time or in other locations. The assumption is that although we may think that a program has worked, we cannot know for sure until we have subjected it to a rigorous test, using controls and large sample sizes. Similarly, what worked in Uganda may not work in Russia but we will learn *why* only if we keep in place the components of a well-designed RCT across these vast cultural and geographic territories.

In EBM-driven Global Health, the ability to evaluate and generalize results across vast differences and contexts of culture, geography, politics, economy (and one could go on) is thus not a problem but an advantage. Seemingly insurmountable differences in contexts and circumstances are the reason to rely on things like RCTs. However, determining whether or not an intervention has a measurable impact is never a simple task. Beyond obvious concerns over which outcomes should be measured and for whom these outcomes matter, social scientists have offered numerous critiques of so-called successful health interventions, noting that although successful by some measures, they often produce unintended consequences that render their overall success ambiguous (Justice 1989; Ferguson 1994; Nichter 2008; Biehl 2008). In fact, much of the corpus of social science research in international public health has been specifically devoted to this topic, and, in the end, much of this work has been used in international health planning in order to revise and re-

formulate intervention approaches (from the work of George Foster all the way to Paul Farmer).

Generalizability is also a tricky epistemological undertaking. Outcomes can be measured in a variety of more or less contested ways, but in EBM an outcome is generally considered valid only if it meets very specific criteria: if it has an established comparative case (using a control or comparative group); if it has statistical power (a large enough *n*); and if it can be shown to be reproducible in other settings using the same standard methodology. In other words, the logic of the RCT uses a set of criteria that are peculiar to itself in order to establish validity. Outcomes are tautologically measured by one criterion above all others: namely, by the degree to which the intervention is designed as a randomized controlled trial; or, failing that, by the degree to which it has *some* design components of the gold standard. Thus generalizability, which is established by following standard methodological protocols, becomes one of the most important determinants of the value of any particular health intervention. Not surprisingly, the validity of context-specific interventions that demand methodological specificity is apt to be questioned. That is, regardless of how well any particular intervention seems to work in a specific location (e.g., giving out free condoms in rural Nepal to prevent STDs), the notion that a local success is a positive outcome is supplanted by the notion that unless it used the controls, randomization, and statistics that guarantee generalizability, it is of minimal usefulness. Thus, a study that compares the distribution of free condoms with the use of social marketing strategies to sell condoms to prevent STDs in neighboring villages is seen as much more useful than a study that simply documents lowered STDs in a village where free condoms have been distributed. But a study that compares two methods of distribution and then randomizes these distribution methods across multiple villages and generates enough individual outcomes on STDs to compare villages would be the ideal method for establishing what really works to reduce STDs in relation to condom distribution.

What is lost in this leap to the use of rigorous statistical methods is the recognition that sometimes context itself determines what methods can and cannot be effective. The conditions of life and living in the places where interventions are offered often make studies that have all (or sometimes any) RCT components extremely difficult, if not impossible. Using RCT as the standard and then trying to deploy RCT-type interventions

in places where they are not appropriate sometimes means ignoring the unforgiving realities that inhibit ideal RCT design. Other times, it means giving up on such designs altogether and accepting that the outcomes of one's intervention may never be taken seriously in global health publication circles. Thus, for instance, in the hypothetical case above (of condom distribution), problems of "contamination" between distribution villages either have to be addressed (making the RCT impossible) or ignored (making the RCT flawed). If one did distribute condoms in a village and noted declines in STDs, it would be difficult to make the case that this observation had scientific value. It would merely be a subjective observation until one added in measures of comparison, control, randomization, and large " n "s that can be powered.

The challenges that face attempts to implement RCTs in global health go beyond this set of concerns. All kinds of empirical realities may impinge on the ways that RCT data is produced, and they may condition the outcomes that one sees. Questions about the social and infrastructural dynamics that get involved in managing the distribution, consumption, and use of condoms (to continue with the hypothetical case), and the meaning of cash versus gift transactions in relation to notions of sexuality in villages (based on religion, family pressure, sentiments of love, and the like)—and these are just examples—might, if pursued, lead one to the conclusion that reducing STDs has nothing to do with how condoms are distributed, but rather with things like the religion of the person distributing them, the color of the packaging, the commitment to healthy families over and above the cost of condoms, or the social and behavioral obligations that come along with receiving gifts from a health center. All of these considerations may be vital to explaining rates of STD reduction, or the lack thereof, but unless the RCT study is designed to include them all, there will be no way to know this, and assuming that the RCT has captured the essential variables that matter may lead to misleading claims.

While RCT-research advocates argue that any and all of these potentially confounding conditions could be adapted to, or incorporated within (or somehow managed away by), well-designed RCT planning, what instead happens is that such confounding variables are often treated as "static" in the system and ignored. What really matters is getting sufficient numbers. In an RCT numbers seem to carry intrinsic validity. If there are sufficient numbers, this by itself lends the study a perceived reliability that

transcends questions about how those numbers were produced and what they actually stand for. Any notion that context-specific, tailor-made, and nonreproducible interventions and findings might just be the most efficient way to conduct international public health interventions falls by the wayside as generalizable results formulated around the RCT standard take precedence.

Approximating the Ideal: EBM in Public Health

There is a history of controversy over the marriage of EBM with public health. Even within countries like the United States this merger has not been easy, and many of the ongoing debates are pertinent to efforts to understand the globalization of these methods. Public health scholars and researchers have raised concerns over the quality of EBM as implemented in hospital clinical care and over the validity of its evaluations. In fact, over time, the notion of the “rigor” of EBM in public health has undergone some changes. Whereas the best evidence-based *clinical medicine* was seen as relying on randomized controlled trials of medical treatments, “evidence-based public health” (EBPH) was initially thought to be capable of achieving only an approximate version of EBM, and not necessarily rigorous RCT work. Notably, it was thought that the link between intervention and outcome might become visible only after a longer period of time had elapsed than would be allowed for in a controlled study. It was noted as well that in most public health worlds, multiple disciplines were involved in an actual intervention, thus complicating randomization—among other things (Brownson et al. 2003; Briss et al. 2004). Initially, what proponents of EBM in public health called for was a “quasi-experimental” design. They acknowledged that interventions often lacked “true” comparison groups (Brownson et al. 1999). Journal articles sought to promote standardized ways of assessing public health interventions and of assessing the literature produced about them, so as to make the best of what was implicitly seen as a messy situation (Briss et al. 2004). Granted, real-life public health scenarios did not look or feel like the laboratories or the clinics where EBM techniques were born, but, with effort, it was felt that they could be made to approximate these research spaces, conceptually and epistemologically.

It is important to remember that one branch of public health that is equally important in international health already relies heavily on the use of statistical methods: epidemiology. Long-standing debates between social scientists who study or use epidemiological methods had aroused some awareness of the pitfalls and limitations of using statistical methods to account for complex social phenomena (Janes, Stall, and Gifford 1986; Susser and Susser 1996; Treichler 1999; Trostle 2005; Shim 2005). **Researchers noted that the use of statistics at the population level could serve both a political function (in advocacy) and a depoliticizing function (by rendering social problems “objective” facts).** They also argued for methods that mixed a variety of strategies to create more complete pictures of outcomes, including ethnography, rapid assessment, and case studies. Some have documented how the use of epidemiological models negatively impacts clinical caregiving (see Gifford 1986 for an excellent example). However, these critics did not anticipate that the uptake of RCT-style research in EBPH might colonize all other research, under the mandate that all international public health interventions should look like experiments. The use of statistical models in public health work is complicated by the adoption of RCT as the gold standard, as researchers of all kinds are asked to embrace the idea that in order to be scientific their interventions must be designed like clinical trials. This goes far beyond using statistics to make strong analytical claims; it requires doing research to get statistics in a whole new way. At its worst, the advent of such experimentalist platforms for research has made the job of producing statistics as important as, if not more important than, the actual medical goal of the interventions they hope to implement.

Over the past decade, it has become increasingly imperative for public health efforts to front-load their efforts with more rigorous and scientifically competent design strategies. Although this is not yet universally true, the messy world of public health has begun to look, here and there, like a bench science laboratory. Today, it is common to see public health interventions that use randomization with comparison or controls as part of their delivery strategy. Public health epidemiologists have, at times with celebratory zeal, taken the basic four-phase RCT model designed for pharmaceutical research and adapted it for behavioral and public health interventions. Increasingly, the earlier “quasi-experimental designs” that preceded EBPH have lost legitimacy, and the literature based on such designs

has been judged of limited use (Briss et al. 2004). Simultaneously, public health has witnessed a ratcheting-up of efforts to make all intervention studies conform to the rigors of phase I-, II-, or III-style research interventions.⁹ The shift to using an experimental platform for public health interventions is now considered not only possible, but preferred.

Many contemporary journals of public health (the *International Journal of Public Health*, the *WHO Bulletin*, the *International Journal of Health Services*) and the standard clinical journals (*BMJ*, *JAMA*, *NEJM*) are filled with public health research publications based on RCT designs. Titles often include the phrase “A Randomized Controlled Study of [fill in the blank].” In these journals and bulletins, “research” articles are designated as such on the grounds that they use this study design, while other articles (based on observations, social behavioral analysis, etc.) are identified as “news reports” or “editorials.” The latter are not “research,” and are thus treated as if they are of questionable reliability when it comes to evaluating health care achievements. Described as “qualitative studies,” these “soft science” pieces are treated as fun stories that are empty of real scientific value. This is true despite the fact that such findings are often based on rigorous empirical methods, just not the methods of the RCT. Whereas a mere ten years ago, one would be hard-pressed to find a single published article that used an RCT study design for a public health intervention, such studies are now commonplace. The RCT imprimatur seems the most certain route to publication in most of the reputable journals that deal with either international or public health.¹⁰

What is important to recall in all this is how dramatic the transformation has been. In the past, international public health publications typically offered straightforward impact reports. How many children were immunized? How many people were using contraception? How many clean water systems were built? How many hygiene programs were delivered through outreach? Obstacles to interventions, or interactions between one health problem and another, were presented in articles that described the complex local context of global programs and the reasons projects worked or failed.¹¹ These simple but obvious, and often context-specific, ways of addressing efficacy were sufficient for many years and for many people. This was particularly true in the small NGO world, where reports were generated for donor organizations in ways that highlighted the specific outcomes and inefficiencies of a target program or community.

Nowadays, similar efforts are seen as woefully deficient. Labeled “reports from the field” or published under the heading “News,” these simple counting exercises are seen more as an attempt at public relations than as an approach to producing compelling scientific results on which to make claims about outcomes or about the benefits of one type of intervention over another. Meanwhile, a different kind of study is called for when facts about the impact and effectiveness of programs are at issue. Global public health interventions must show that they do more than simply count the number of immunized babies, safely delivered mothers, diarrheal deaths prevented, or hygiene and nutrition lessons taught. An intervention must demonstrate that it achieved results because it was designed as an experimental study to begin with. These results can be useful only if they show that they were better than chance (randomized), and better than nothing (controlled), and, above all, generalizable.

Randomizing the Whole Country: What Counts in Counting Health Outcomes

Thus global health emerges today as an experimental research endeavor. Glancing through journals, we find that everything from “rates of diarrhea” to the “use of antiretrovirals” to “the benefits of combining HIV treatment with contraceptive planning” is now being investigated in randomized controlled research trials.¹² Researchers must give some interventions to some people and either something else or no intervention at all to others, and then they must make sure that these groups of people don’t cross-contaminate, and then that their subject numbers are large enough to show significance or, as I was told by a program officer with regard to my own research (to which I will return below), enough to do “a good power calculation.” If it is possible to run statistical analyses and power calculations on the results . . . then *voilà*: good science and (therefore) good global health.

An exemplary case of this was shared with me by a colleague. It involves a public health policy project being conducted in an island nation. This study, which I will call the Island Insurance Study (IIS), focuses on the impact of health insurance schemes on child health. The intervention is grounded in politics that are, to my mind, laudable: it is designed to

show, with measurable outcomes clearly hypothesized from the start, that the government (and its various lender nations) should view providing health insurance coverage for basic care as an “investment” rather than a “cost,” and therefore as worth the expense. Specifically, the IIS aims to show, using RCT designs, that paying for health insurance is not only cost-effective but beneficial for the health of children. IIS has been implemented in all public health and clinical-care centers across the nation and scientifically designed with three arms: (1) an intervention arm in which expanded insurance coverage is provided, (2) an intervention arm in which providers are compensated at higher rates for better care, and (3) a control arm in which clinics and patients receive neither benefit. The intervention is designed as a research project in which, as the primary researcher on the project proudly announced to me, “the entire country has now been randomized.” It is a spectacular achievement.

In the IIS project, outcome measures are based on physiologic and cognitive assessments, derived from data collected at all of the clinics. Although clinical services are provided throughout the country, these services are being reconceptualized specifically in order to enable the creation of an experimental platform for research, with each kind of care delivery serving as either control or contrasting arm. That is, no health care will be provided unless it serves as a tool for data collection. The project is referred to by its author as an “experimental evaluation” of the policy impacts of these different forms of health care provision. It is simultaneously a study that will demonstrate the value of evidence-based public health research in creating evidence-based public health policy. Finally it will provide valuable insights about which insurance strategies produce the best measurable health outcomes for the public. The deployment of EBPH thus has interesting implications in that it enables interventions to serve two masters simultaneously: both to evaluate health outcomes and to build and sustain a research infrastructure.

Intervention-cum-research projects in global public health often use the notion of a “natural laboratory,” suggesting that the social world can generate laboratory-like data. Often, outcome “measurables” are seen as the end product of the intervention as much as the health outcomes themselves. Good data, it would seem, is as valuable as (in some studies perhaps more valuable than) a positive health result. The data that is particularly sought after is the kind that can be used to generate statistically robust

calculations and comparisons that, once analyzed, can tell the truth about which interventions are most effective at a population level. Often, this means creating a research platform that eliminates the need for data collection about complex social realities, and using select variables as proxies that will tell us indirectly about these larger and complex social realities. For example, in the IIS project, the intention is to use an RCT design to make claims about entire populations, but the evaluation indicator most readily available is children. Because children are seen as an easy target for data collection, they are being used here as proxies for everyone else in the system. Similarly, the study selects “tracer” conditions for measuring physiologic outcomes—proxies for general states of health and diseases. In this case, the tracer conditions are pneumonia and diarrhea (with a wide spectrum of biomarkers). To measure cognitive outcomes, a battery of intelligence and psychological tests are used that assess well-being and the “normal” development of children. Being able to show that some children have better physiologic and cognitive outcomes than others will be, in other words, a positive result for the intervention, and the intervention is therefore designed around an imperative for measurables—data sets that can be obtained in sufficiently large numbers to ensure some degree of reliability. In order to get good numbers, all sorts of proxies must be used. Reality is best captured in the RCT in small measurable bites rather than in its vast and unmeasurable complexity.

What is interesting to me about the IIS intervention is that right from the start it displaces old straightforward questions about what justifies projects like this. While it is clear that the project hopes to improve the health of children, and that it is noble in this effort, it is not clear to me that the IIS design is the best way to do this. Why not roll out insurance schemes across the country and then compare health statistics ten years down the road? It is a large and potentially messy way to generate data, and it won’t be based on use of controls, but it might show dramatic health outcomes anyway. Doing an RCT instead is justified on two grounds: first, to establish an experiment that will show, definitively (in the terms of EBM), what kind of insurance helps most, and, second, to establish that this form of research is possible in large-scale public health projects. My point would be to note that interventions designed in this way actually change the kinds of questions that can be asked. If ten years ago an intervention would have been designed simply to see whether children would

benefit from having health insurance by counting reported diseases and collecting morbidity and mortality numbers, today's intervention must do much more than this. It must be designed to show difference in outcomes in different populations based on the variables of the experimental design of the intervention. Not only are children allowed to be proxies for entire populations, but some kids will have to show that they do more poorly than others. We will have to accept numerous premises concerning proxies and control groups that may raise questions about the research design, but at present this model for research is seen as, again, the gold standard. It is useful thus to explore what else gets produced in generating truth about health and health care in this particular way. Each and every intervention goal is recalibrated to meet the needs of the statistical model, while other possible ways to get the sought-after truth of whether or not insurance helps keep kids healthy are potentially pushed off the table.

In addition to evaluating the benefits of one kind of intervention versus another or versus no intervention, the IIS, like other RCT interventions in global health, will also attach itself to the partner-in-crime of EBM: cost-effectiveness. That is, in the new EBM of Global Public Health, economic calculi are affixed to interventions in ways that make cost something that must be accounted for in the same ways that things like "rates of diarrhea" are counted (and studied statistically). This makes sense, because one wants to show that spending money on health care (however much or however little) "pays off" in health dividends in the end. But this tethering of health to cost variables constitutes another way that EBM is changing global health and how we go about trying to help people achieve health.

Not surprisingly, cost-effectiveness is not the only fiscal concern of the IIS project. The IIS, like other global health interventions today that strive for EBM standards, has to spend a lot of money on data collection, processing, and analysis. The data processing for the IIS study, involving the massive use of biomarker and cognitive assessment instruments, large-scale data-entry teams, and epidemiological biostatistical analysis, is being undertaken by a locally based subcontractor. The intervention itself will help stimulate the local economy, which is surely not a confounding variable, but it is a fact that points to important new demands that are being placed upon anyone hoping to do Global Public Health in ways that will be recognized as scientific, useful, and valid.

My point, as an observation more than critique, is that the experimentalization of the entire clinical health-service system in the country where the IIS intervention is being done is an accomplishment that is not simply the product of a desire for reliable and effective health care delivery. It is also the product of a convergence of specific neoliberal political and economic trends. By neoliberal, I refer to the normalized privatization of public institutions, in which creating pathways by which public-sector activities can be transformed into private-sector, profit-garnering infrastructures is the norm. In this transition, the problem of accountability to a market becomes a nonnegotiable priority. The effort to roll out and scale up use of EBM in Global Public Health comes as a package in which the management of cost (of intervention as well as of doing the research) is inextricably sutured to the ways the research/intervention is designed.

This suturing of fiscal to scientific kinds of health data in and through the research enterprise shows up in the new medical industries that come into being in countries where “scaling up” research and biopharmaceutical infrastructures is a priority (Sunder Rajan 2006). It results in the capitalization of health-sector research and in a consequent tethering of health to biological models that can deliver high-cost research opportunities. It also shows up in more subtle ways in the global health world: as when interventions themselves must incorporate *labor-* and *cost-*intensive research measures. Finally, the impact of neoliberal trends toward privatization of the health sector shows up in the presumption that the best public health must be ever-attentive to the question of cost-effectiveness—that is, to insisting that economics and sustainability are a health priority (as opposed to recognizing that some things just cost a lot and can’t pay for themselves, or that they will require a long-term commitment before they achieve self-sustainability).¹³ Thus, even though the IIS intends to show that a *government* health care insurance program will improve health, it is deeply committed to working within a US model of entrepreneurial research that is rapidly privatizing and creating for-profit infrastructures for health research in and through commitments to Global Health.

The shift I am pointing to is one in which fiscal concerns have become a foundational, and perhaps tyrannical, presence not only in what we study, but also in how we study it. For the sake of efficiency, projects like the IIS are now required to work with large corporations because they are capable of processing voluminous quantities of data, for a price.

Simultaneously, research-based interventions are asked to consider cost-effectiveness as a variable that penetrates and pervades all that is done. These are, I believe, outcomes of the adoption of neoliberal ways of doing things in Global Health, and they are tied to the methods of EBM and RCTs.

A prime example of what happens when all health concerns are tied to a market or economic model is the use of DALYs (Disability-Adjusted Life Years).¹⁴ Designed to measure the impact of illness and disability on a community and to help prioritize investments of health intervention aid, DALYs are a tool for gauging a sociological phenomenon (quality of life) on a quantitative scale (according to the value in productivity of that life). That is, DALYs equate quality of life with a fiscal criterion for the value of life. Critiques of the shortcomings of DALYs are plentiful. Nichter (2008) summarizes them well. Still, few note the seamless way in which the insertion of such metrics into the messy world of international or, now, global health lends credibility to the idea that everything (from the impact of disease to the impact of an intervention) can be measured quantitatively. This notion has emerged from within a nexus of neoliberal priorities that privilege fiscal bottom lines above any and all other possible bottom lines. To see how the institutions of evidence-based research further contribute to this particular form of neoliberalism, consider another project.

Evidence-Based Medicine in Global Public Health: Producing and Privatizing Profits

In 2008, I witnessed an effort to bring a resiliency-training program to low socioeconomic status (SES) school-age children in a large US city. The public school systems in this city are among the least successful in the nation, judging by performance measures, and this is attributed to such factors as entrenched poverty, high levels of violence, limited to no social or health services within the schools, and high rates of ongoing trauma. I first heard about the desire for this project from a city resident who had worked in the schools and had seen the children's needs first hand. In response, I joined a collaborative partnership that included a non-profit resiliency-training organization that had done extensive work with children, and a research team that had experience working with NIH

funding in large-scale intervention research programs. The team that came together to design the project was headed by a clinician-epidemiologist who was joined by the nonprofit director, a well-known psychology researcher, and a well-established anthropologist (myself). The team's epidemiologist leader set the tone of the research and intervention strategy. The first step would be figuring out the best way to achieve an effective randomization strategy. A control group was needed, but the team did not want to enlist schools or children to participate if they received no intervention. The design eventually used a staggered intervention strategy so that one group could be used as a control initially, even though they would eventually get the intervention. This was a reasonable solution to an ethical conundrum: how to make the research design robust but at the same time avoid denying participants the potential benefit of participation. It is a conundrum that is as serious for community-centered global health intervention studies as it is for hospital-based clinical trials. Of course our study included enough students (a large enough n in each group) to generate reliability and "statistical power."

The second step was to identify *measurable* outcomes, using questionnaires to collect data on a wide range of variables: included were the subjects' development of internal and external psychological assets, their mental-health status, behavioral changes over time, and school achievement. Some of these outcome indicators would be available from the schools (via test scores, attendance records, graduation rates, and rates of expulsion, suspension, and disruptive incidents in the classroom). Other indicators, however, would have to be derived through data collection methods that would involve the use of questionnaires and pre- and post-intervention surveys. Our hypothesis was simple: the schools in which we provided resiliency training would have better outcomes on all of these measures than those that did not.

As the anthropologist on the project, I assumed at first that part of our research would involve designing questionnaires for assessing such things as psychological or mental health that were culturally appropriate and specific for this target population. Much to my surprise, I learned that many of the assessment instruments we would use were instead standardized and already in use internationally. Not only would there be no need to tailor questionnaires to local cultural conditions, but the very idea of doing so was anathema to the rationale for using questionnaires

as instruments, which was to elicit objective information about psychological states. Tampering with the questionnaires would undermine their utility. I also learned that these instruments had to be purchased for a license fee and a cost-per-questionnaire fee from several companies, one called “ASEBA” (Achenbach System of Empirically Based Assessment) and one called “PsychCorp.” We hoped they would offer us the “research-use” rate, which was about half the commercial rate, and would amount to roughly US\$1 per questionnaire per child. We also had to consider whether we would need to purchase the scoring programs and data-entry services that the companies offered in order to process the data once we had collected it.

Instruments of research that rely on massive data-collection techniques are themselves a by-product of the notion that reliable evidence comes only from research projects that use statistical methods to document all of their results. EBM, which relies on these techniques of truth seeking, is committed to making the world a better place by collecting more reliable information about what can be known in the real world of personal experience (like the information we hoped to collect using these formal assessment instruments). Along the way, EBM also promotes economies of scale and regimes of truth that introduce a new kind of fiscal profitability to the world of science. If our results were to be reliable and we were going to be “objective,” we would have to buy the assessment tools that PsychCorp and ASEBA were selling. (To put it bluntly, these companies own and are in the business of buying and selling reliability and objectivity.) If we used their instruments, our methods, and therefore our results as well, would be standardized, comparable across not only the schools in our study but, in the ideal imaginaries of RCT advocates, comparable across the entire globe.¹⁵

The logic of “market-driven research methodology” rides on the coattails of an already commodified health care intervention industry. The notion that intervention instruments constitute private property, protected by intellectual property laws and patents, has long been accepted and is key to the market achievements of modern medicine in the developed world. What we are witnessing now is the further migration of this privatization and commodification into the world of social science and health development research. It is no longer only the treatments (drugs and technologies), but also the research and intervention instruments that are seen

as patentable private property that needs to be retained by its owner until it is paid for. All of this is augmented by rhetoric that insists on RCT as the unequivocal “gold standard” for truth. Federally supported research projects of this kind do not follow a straight path to serving the public. Rather, they first serve to feed and fuel a growing number of private-sector companies whose main business is to generate instruments and results for evidence-based research interventions.

I note that many researchers and public health advocates stand in an awkward relationship to this economy, frustrated that they have to take part in it, while at the same time feeling some sense of urgency to participate as entrepreneurs in the interest of their own work (which is itself threatened by fiscal cuts in public support for higher education). However ambivalent the feelings of some of the participants in this research economy may be, it nevertheless comes as no surprise to learn that the principal investigators on many projects are also the heads of private-sector research companies that handle subcontracts for data processing and consultation for other interventions in other locales. Universities are providing seminars and training in how to make our research more entrepreneurial and how to develop spin-off companies from our work that might attract investment capital. These new public-private partnerships, as they are often called, fuel a sense of win-win for research enterprises and for researchers alike, but how well this model will attend to the needs of the global poor is not yet well understood.

In global public health, conflict of interest is not yet regulated the way it is in the biotech academic world, which monitors and attempts to regulate the degree to which researchers can profit from the interventions they are studying. Although such conflicts are supposed to be disclosed for projects in the United States, the simple fact that an investigator is earning money is rarely sufficient for a “conflict of interest” judgment. The use of assessments deemed “standardized” instruments in an investigator’s area of expertise and simultaneous involvement with the companies set up to make those instruments available is sometimes unavoidable.

In our own project, we figured that we would need at least five assessment instruments, most of which would have to be purchased from private-sector corporations.¹⁶ Only a few of our measures could be obtained via public institutions, such as the CDC or the school district, which held records of nationally standardized assessment tests. In sum,

our research would be made “credible” and “fundable” only if we agreed to push it through these required passage points of commerce—in and through the private and for-profit infrastructures of EBM—to the corporations that had begun to emerge as profit-driven subsidiaries of the Global Health World.

The NGO-Industrial Complex and Audit Culture: EBM's Profits

There are many kinds of evidence, and any and all of them could potentially serve different public- or private-sector interests. Nonetheless it is clear that in the push to privatize and to create public-private partnerships, some kinds of evidence are seeing more frequent use than others. An anthology edited by Marilyn Strathern (2000) explains the *historical transition toward more evidence-based forms of accountability as an outcome of new infrastructures of “audit culture” tied to late twentieth-century neoliberal reforms*. The reduction of complex political, social, and health problems in developing nations to numerical data about things like GDP, poverty rates, and infection rates has long been a stated goal of international aid organizations (Harper 2000, on IMF; Ferguson 1994). Such practices are highly ritualized and culturally normative. And, in complement to what Herzfeld (1992) identified as a rising *bureaucratic indifference* that results from the imperative of the audit, and to Ferguson's point that such practices inherently “depoliticize,” it is important to focus also on the ways in which emerging forms of audit are highly productive in their own right. That is, instruments of audit are productive in more than the sense that they produce “effects,” and sometimes unintended “effects.” They are productive in a fiscal sense as well. While some things are effaced and erased by the audit (like politics and unmeasurable successes and failures), other things are made more visible, and most of these can be made fiscally profitable.

The rising demands of “audit culture” bestow a perhaps undue importance on EBM in the world of scientific medicine. En route, these demands have fostered whole new evaluation and assessment industries, which, in turn, have engendered more and more private-sector companies devoted to generating the kinds of evidence and results demanded by publicly

accountable evidence-based medicine and public health. Accountability and evaluation NGOs, for example, are now plentiful, a vigorous growth industry in the world of international health development, and they are altering the contours of global public health (Adams et al. 2008). In the US city where we had hoped to work, I was told that Save the Children paid a private assessment firm something on the order of a quarter of a million dollars to create a report on the impact of their own resiliency trainings. Publicly funded research teams are increasingly compelled, by the litmus of the RCT, to use these research instruments and assessments in order to establish and affirm the effectiveness and impact of their interventions. They have become widely recognized measures of reliability and generalizability and so also serve the purpose of demonstrating *accountability* to donors. These are the pillars upon which audit culture is built. Although some audit instruments are publicly available, many are privately owned, and this is consistent with neoliberal assumptions that private companies are *more accountable* than government-run organizations. Private-sector, for-profit, and nonprofit businesses are, in short, the new arbiters of truth and also the new technologies of profit in evidence-based global public health.

An uptick in the proliferation of NGOs in the world of international health can be dated to the 1970s, along with initial neoliberal and structural adjustment policies that tried to put health development into the hands of local NGO facilitators. What we are witnessing now is the birth of something I have heard called an “NGO-industrial complex” in global public health. At least one arm of this industrial complex is fueled by the rising demand for assessment and evaluation services that define the benefit, or lack thereof, of interventions that now have to come packaged as research projects. Like contract research organizations (CROs) in the pharmaceutical world, some NGOs in the world of public health garner high-ticket contracts for private and public research and intervention, and thus operationalize the business of public health as a private-sector enterprise. NGOs come in all sizes and shapes. Some are for-profit, some nonprofit. Some perform their own interventions; others contract to assess the interventions of other NGOs. Not all NGOs are involved in audit, but audit NGOs and companies that offer services for hard and fast data crunching are proliferating at a very fast rate, and they are being paid in many cases by NGOs that have historically been commit-

ted to humanitarian interventions that made only minimal investment in experimental designs.

Two interrelated trends emerge at the present juncture. First is an increasing reliance on experimentally designed interventions as the new gold standard for intervention programs, specifically the RCT model for intervention research. Second is the increased privatization of public health interventions in and through private-sector research groups, NGOs, and companies that offer audit, statistical support, and research design support—a trend that uses multilateral, bilateral, and government funding but responds to private-sector accountabilities. Evidence-based medical standards, and the need for experimentalizing the field of public health, are two components of the shift that enable this privatization and the proliferation of new fiscally driven intervention models.

In Global Health, the transition to more “evidence-based” research is mapped onto a shift away from liberal concerns with preserving public institutions devoted to promoting health over profits and toward neoliberal concerns with promoting private-sector infrastructures that tie health to market accountability. This shift is negotiated in and through the experimentalist rubric of EBM that uses the RCT lexicon of “assessment,” “evaluation,” and “generalizability” as an essential component of intervention. This is true whether the intervention team operates under the umbrella of public health institutions, academic institutions, or either nonprofit or for-profit NGOs. Despite a long history within the liberal health sciences of quantitative approaches to scientific evidence designed to *protect* health care as a public good (Townsend and Davidson 1982), many of the instruments of today’s quantitative science have increasingly aligned with the rise of private-sector investment, leading to a disenfranchisement of both the public sphere (institutions of government) and populist public-sector interests (in health over profits, for example).

Today, evidence-based methods, and particularly the use of these in experimental research interventions that include some measures of cost-effectiveness, by a research team that uses high-cost, high-tech methods of data collection and analysis, are deemed more reliable than other methods that have long held sway in global health. What is not often noted is that this shift brings with it a different kind of accountability. Today, accountability to a market is fused into the heart of the research

methodology itself. Not only must researchers show that they are responsible in data collection by spending money on these cost-effective instruments of data collection, but NGOs who often deliver health care are asked to give proof of their market sustainability as private entities. Everyone who wants to get involved in health implementation these days must, it seems, get into the business of assessing interventions, if not health itself, primarily on the basis of these techniques. The turning-over of intervention work to measures driven by the market tends to reduce the work of “experimental research” to questions of fiscal accountability, and RCT models of research help this process along.

Invalidation and Commodification: EBM's Erasures

What sorts of things become invisible as intervention in international public health shifts its ground to occupy and advance an experimentalist platform? In 2005, I documented the ways in which evidence-based medicine derailed a project in which I was involved that aimed to train health workers in techniques of safe motherhood in rural Tibet. Our project, funded by a US government agency and approved by the Tibetan Autonomous Region (TAR) Health Bureau, was nearly ready for launch when we were informed by our funders that our methodology was insufficiently rigorous. Our “numbers,” it seemed, would be too low to enable good research. Despite the fact that more women died in childbirth in rural Tibet than in just about any other country (Adams 2005), and despite the fact that our training would be conducted in a large rural prefecture (with roughly eight villages) and was endorsed by the TAR Health Bureau, we were told that in our catchment area “not enough women die to get a robust power calculation.”¹⁷ We would not be able to tell whether our intervention resulted in a reduction in maternal mortality or whether any such reduction was simply due to *chance*.

In order to keep our funding, we were advised to work with a research consortium NGO based in Maryland, which could help us restructure our project so as to render it “more scientific.” Our funding administrator referred to our need to generate results that could be generalizable beyond the Tibetan context. Our Maryland NGO research consultants advised us to abandon the training for safe motherhood project and to focus instead

on infant mortality (because the numbers would be better and because geography prevented our extending our study to a significantly larger area to get more women in our sample). Since we did not have in hand a training program focused on infant diseases, however, and since we had a commitment to the TAR Health Bureau to work on safe motherhood, we had to be more creative. In the end, we agreed to undertake a randomized, controlled clinical trial comparing the efficacy of traditional Tibetan medicine with that of a Western drug that prevents hemorrhage in delivering women. The study would be conducted in several hospitals in Lhasa, and it would involve a large team of data collectors, randomization, and controls. Meanwhile, the people on the team who had committed to helping the Health Bureau train rural workers in safe motherhood splintered off from our project and set up a separate NGO to do this work with private funding.

Although the investigation we pursued in Tibet had little relevance for the Health Bureau, and in fact delayed the safe motherhood intervention until a different funding source could be found, the *research* project was successful. It was a negative study, in that we were *not* able to show that Tibetan medicine worked better, but the project was considered successful to our funders in the sense that we were able to do an RCT. We trained a small cadre of Tibetan and Chinese health employees in methods of reliable statistically based, IRB-approved RCT research. The most compelling criterion of success was thus the research, and not the cure per se, or even the health outcomes. We were showcased by the US funding agency for having set up a Tibetan IRB at the Lhasa Health Bureau. Our impact on Tibetan health concerns was, however, decidedly mixed.

Evidence-based medicine has become a screening mechanism, a sieve through which truths about health, and judgments about the effectiveness of health interventions, must pass. EBM has also become an end in itself. This sieve/end point enables funders to shut down projects or keep them alive based on how strictly they conform to the model of best-practice EBM and EBPH research. EBM in all its forms becomes conveniently linked, as an apparatus of audit, to the growing NGO-industrial complex—the subsidiary growth industry in public health infrastructures that is being turned over more and more to the private sector.

Petryna notes that “clinical research is now a worldwide data-making enterprise” (2009:187) that spills far beyond the contours of the phar-

maceutical sector. It generates growth coincident with the growing need for more scientifically rigorous models of health intervention. But it also “weeds out” other kinds of activities and resources. Contract epidemiology firms and assessment firms that get bids to go in and “evaluate” school resiliency-training programs, or to measure the impact of public health interventions, or to do the data processing for large public health RCTs, are often in the position of hoping for outcomes that may differ from those most often prioritized by public institutions (and I would add some humanitarian aid institutions as well). Research firms are often more interested in getting results that show the study was done well, as opposed to caring primarily about whether the outcome of the study actually improves health. As a result, other evaluation and intervention methods that have long been used may be marginalized or discarded outright. Interventions of the kind that cannot be subjected to experimental evaluation become not obsolete but seen as somehow of poor quality, as experimentally designed interventions become promoted as the gold standard.

If doing EBM in Global Public Health changes the kinds of questions we ask in our research as well as the way we think about interventions and health-delivery efforts, then it also changes how we think about outcomes. Other kinds of outcomes and intervention strategies that do not meet these standards tend to fall by the wayside as if they can’t tell us much about anything. Some possible interventions are made to seem useless and cavalier from the perspective of using an evidence-base; you can do them, but you can’t really know for sure that they help. This includes things like paying for the training of rural health workers so that they can learn safe delivery techniques even though we cannot prove that their interventions work better than chance; paying for government health care insurance even though it cannot be shown to reduce morbidity using the limited set of measures in the RCT-designed intervention.

What happens to populations whose health needs are not ideal targets for experimentalist research and intervention projects? How does one even show that such interventions are vital to health and survival when they can’t be measured using RCTs? What and who will attend to the needs of those for whom one cannot find “generalizable” or comparable health solutions? Questions about who evaluates the evaluators, and who audits the auditors, are seldom asked. Concerns that don’t fit

into RCT-style research tend to drop off the radar, become relegated to an insignificant backwater of public health because they cannot be made to fit into the mainstream of evidence-based public health. Despite the optimism among EBM supporters who argue that any and all of these things could be studied using experimental research methods, the reality is that the bar of RCT is simply too high for many lived realities and determinants of health in most of the world. And then, there are all the things that simply can't be counted, or that, even if they could be counted, in being counted would be turned into misrepresentations of their true relationship to health. What I am arguing here is that there are unintended outcomes involved in reprioritizing health goals so that they meet the EBM standard for research interventions. I would ask: Does this growth industry actually assist or impede the work of caregiving?

Within the world of international health development, many of the people involved in implementing and evaluating programs are aware of the contested nature of evaluations. Patton (2004) noted growing concern over the standardization of monitoring and evaluation procedures in HIV/AIDS research and warned that using experimentalist linear methods of evaluation poses numerous problems. He observed that the needs of local operating clinicians and health providers often differ from the needs of a research program: "the data most useful to local programs fail to provide what international donors want" in the form of global comparative data sets on outcomes. He concluded, "Donor reporting and accountability demands burden the system and engulf local data collection efforts to such an extent that evaluation is little used for program improvement." Most evaluation strategies obscure the stories of real people and real-life circumstances in favor of numerical facts. Patton's greatest concern was over the use of linear, quasi-experimental models of evaluation:

The dominant concept of evaluation . . . is a traditional inquiry into an autonomous intervention delivered in a linear-outcomes model to isolate and attribute its impacts, including conducting quasi-experimental designs. I am openly skeptical about the utility of such designs in these circumstances. Indeed, the controls needed to even attempt such designs risk having the evaluation design interfere with the effective creation and implementation of complex and dynamic systems interventions that are too messy and emergent to be appropriately evaluated by static designs and linear, mechanistic attribution models. (Patton 2004:169)

The still-emergent sense that Patton expresses, that program monitoring and evaluation must be seen as a complicated task involving the exploration of entire systems, complex specificity, and impacts beyond the catchment of the experimental design, has met with a huge amount of skepticism from many researchers in global health sciences. Advocates of experimentalist-driven research often respond to such concerns with skepticism of their own, noting that much of what gets published and reported in international public health circles tells us little if anything that is reliable. Unless research is designed effectively to provide significant and reliable data, the information it generates cannot be used in turn to generate responsible policy or intervention plans. A good example of this is an article titled “Weighing the Gold in the Gold Standard: Challenges in HIV Prevention Research,” an editorial review in the journal *AIDS* (Padian et al. 2010). The authors looked at thirty-seven published HIV prevention studies and, rigorously evaluating the quality of the research designs in relation to the results, reported that 90 percent of the studies had “flat” results, meaning that regardless of published claims, a study could not show positive results either because it began with a flawed concept (an intervention that was not proven to be useful), or because “aspects of the study design, implementation, or context limited detection of a true effect.” The focus of the article’s critique is the last shortcoming: the fact that most studies to date were designed poorly, meaning they were not up to the RCT standard—a failing that, the authors argue, “squanders resources and costs lives.” Like the tyranny of Damocles’s sword, the RCT hangs over our heads as the standard we must all live up to or, failing, be brought down by.

The Disappearance of Subjects, Relationships, and Caregivers

In scientific discourse, credibility often hinges on the scientist’s ability to control bias and reduce the complexity introduced by multiple perspectives. [Thomas] Kuhn discussed scientific knowledge production in terms of the nature and logic of what is deemed scientific, as well as the “techniques of persuasive argument” offered by scientists. However, Kuhn was ultimately unwilling to question scientific rationality, a skepticism that

underpins constructionist claims. . . . Ian Hacking argues that people and their behavior are interactive kinds, capable of looping effects that change them, individually and as a group, as a result of classification. They can potentially destabilize knowledge categories that are meant to be stable. . . . Because of looping effects, the targets of the social sciences are on the move.

—Kelly Knight (2010)¹⁸

One of the problems EBM faces in the world of global public health is that it needs, in the words of Kuhn, Hacking, and Kelly Knight, above, stable objects, but in that world, most of the objects of health care are neither stable nor firm. They are tied inextricably to things social, political, and cultural; and they are, as Hacking notes, always “on the move.” Things like RCTs use an experimental method to create a sense of certainty and stability around their objects of study, whether in terms of measurables, outcomes, or effectiveness. The RCT method provides an illusory sense that there is firmness and stability in the intrinsically messy social world of people, health, and disease. In a rapidly privatizing economy and in a world where competition over resources is intense, EBM is also seen as the solution to several problems in public health: (1) the need for accountability and assessment (the auditing of interventions, which has historically been absent); and (2) the need for an objective platform for globally comparable research, such as generates objects of intervention and measures of effectiveness that seem to have stability regardless of culture, politics, social hierarchies, national institutional variability, and inequality. But in the effort to create such stable objects and methods, some ancillary institutions and interests are created and fed (NGOs and private-sector research investments and profits), and some things are erased. I turn now to collecting the threads of the things that are potentially erased.

I have argued that there are phenomena in the world of global public health that cannot be measured by the use of EBM’s RCT methods. I suspect that because of Tibet’s infrastructure, population demographics, and level of development, it will never be possible to randomize enough of rural Tibet to undertake a RCT of safe motherhood interventions.

There may never be a way to know if the disadvantages suffered by those “randomized” to control status in the IIS intervention are suffering in ways that cannot be accounted for in the RCT design, or if an inability to demonstrate cost-effectiveness necessarily means that providing insurance is not a good idea. Some of the ways in which underprivileged school children in one US city experience a lack of resiliency around trauma may never become visible if we are required to use an assessment instrument calibrated to the national average. Nor will we be able to tell what impact our “controlling” of some schools will have on the children in those schools; for we are not measuring *that* impact. The problem with evidence-based public health research is that much of what we can learn from those who are the “targets” of medical interventions simply does not fit and cannot be made to fit into the epistemological architecture mapped by the RCT. Variability in the nature of health problems from place to place, from person to person, or indeed in the perception of what exactly is causing people to suffer, are all made to seem insignificant in research designs of EBPH. In some rather important ways, these erasures point to the disappearance of a subject—and to the disappearance of his or her social worlds.

As recipients of public health intervention turn into research subjects (a problem noted by Petryna for those enrolled in pharmaceutical research), their ability to contribute truth to the interventionary playing field is reduced. They are able to speak now only through the fine-grained amplifier of the EBM research model. If the things they say don’t fit the study’s template for responses standardized for generalizability, they are winnowed out in the data-cleaning process. Relationships that play important roles in people’s health, including such contexts as families and communities, are not easily standardized, and therefore they are often ignored or made irrelevant to study designs.

But subjects do matter, and subjects who speak in their own peculiar ways and who are embedded in highly mobile social relations particularly matter. In fact, it is possible that their testimonies should be ranked more valid than what RCTs tell us. Robert Castel (1991) noted this in his analysis of risk discourse in clinical medicine. Early on, he warned about the historical displacement of the notion of a medical “subject” by the notion of “risk factors,” such as are needed for the surveillance of populations. In Castel’s view:

The new [preventive strategies of social administration] dissolve the notion of a *subject* or a concrete individual, and put in its place a combinatory of factors, the *factors* of risk. Such a transformation [...] carries important practical implications. The essential component of intervention no longer takes the form of the direct face-to-face relationship between the carer and the cared, the helper and the helped, the professional and the client. It comes instead to reside in the establishing of *flows of population* based on the collation of a range of abstract factors deemed liable to produce risk in general.” (Castel 1991:281)

Castel’s concern over the erasure of a critical relationship of caregiving under new regimes that prioritize population risk assessments over specific and immediate instances of harm (to self or others, what he calls “danger”) is similar in form to that of the subject and his or her social worlds under the hegemony of EBM and the RCT. Relationships of caregiving foreground the specificity of particular cases and of the kinds of individual needs that are visible only in the face-to-face encounter between healer and patient. Such relationships are variable, and, above all, they evidence a kind of particularity, as medical anthropologists have been demonstrating for many decades now. Context matters, and individuals matter, as do the kinds of variations that emerge within specific social encounters that vary from person to person, group to group, context to context. People and their engagements with health do not inherently or easily form “data points” that can be used as measurables. But many researchers nonetheless find ways to identify the easy data points and hope that whatever factors appear *subjective* and *socially complex* will somehow fall away quietly as the RCT model takes shape in research planning. Even if it were desirable to include all of these subjective and socially complex factors, it would be impossible to take all these variables into account in the streamlined world of EBM RCT work, but it would be equally problematic to assume that because these realities don’t fit the statistical straitjacket we want them to wear, they are therefore empirically invalid as evidence of what works, or fails to work, in global health.

It is difficult to imagine the epistemological displacement that would have to occur in order to generate a research or intervention paradigm that could carry as much credence as EBM or even a part of EBM. I

suspect that a new kind of “community-based medicine” approach might provide a clue in its ability to prioritize the contours of the setting in which research takes place as a means of interrogating and designing research strategy (Malone in Offen et al. 2008; Minkler 2004). Some community-based research builds political issues and variable conditions into its method as an inherent empirical foundation upon which to determine, in an objective way, what may or may not work (Epstein 1996, 2009). Results from one site may or may not be generalizable to other sites. Each problem may demand its own solution. Assessment has to be local and specific, even subjective, and sometimes contrary to the rules of statistical reasoning. Partners In Health’s human-rights approach to health care and the work of many nonprofit NGOs that operate small-scale intervention and aid projects in single locations, single countries, or single communities clearly have impacts that go largely unrecorded in the annals of academic and multilateral work. Research projects like these are, like the subjects they aim to study, always on the move, in Hacking’s sense. They are capable of changing to accommodate more realistic approximations of reality and truth than can be produced through the experiment of the RCT. It is worth thinking about what sort of counterregime might be possible when concerns like this are kept in mind.

It is also worth remembering that even when EBM marginalizes the case study, the anecdote, and the particularity of different voices, this sort of data still matters. And it matters for more than its public relations value. It matters because it tells us how things really work, no matter how nongeneralizable such insights may be. Adriana Petryna’s insight that what we need to do in some sectors of global health is to undertake a “rescue of perception” is apt. We perhaps need to recognize that these spaces, relations, and lives that fall off the maps of RCT research, perhaps constitute a different kind of “evidence” in a different kind of “evidence-based medicine.”

Notes

1. I am indebted to several institutions for funding for the research upon which these insights are based, although none of the funding was directed specifically at exploring this topic, including the NSF, NIH (NICHD, NIA), Wenner-Gren Foundation for Anthropological Research, and the Institute on Global

Conflict and Cooperation (UC). I am also indebted to input from a variety of colleagues whose insights have helped shaped this paper, particularly through their contributions to a previous panel at the American Anthropological Association on Experimentalism (organized by João Biehl and Alex Choby), as well as to colleagues Kelly Knight, Ian Whitmarsh, and Deborah Gordon, and the participants in the workshop “When People Come First: Anthropology and Social Innovation in the Field of Global Health,” Princeton University, March 11–13, 2010.

2. Petryna (2009) has explored this thoroughly in the case of globalized pharmaceutical research, which is directly tied to the phenomena I discuss here.

3. Brownson et al. (1999:87) define it as “the delivery of optimal individual patient care through the integration of current best evidence on pathophysiological knowledge, cost effectiveness, and patient preferences.” They note that it involves primarily the ability to track down, critically appraise, and rapidly incorporate scientific evidence into practice.

4. There are critiques from within medicine about the difficulty of implementing strict EBM practices. Sackett et al. (1996) offer a critical exploration of the compounded nature of this problem in resource-poor settings.

5. Taken from the US Preventive Services Task Force, Agency for Health Care Research and Quality (<http://acc.org/qualityandscience/quality/evidence.htm>).

6. I note that there is an ample literature on evaluating social science and behavioral research (Giacomini et al. 2000).

7. The editor directed the author to a reference for producing high-standard qualitative research, noted above (Giacomini et al. 2000).

8. This point is made by Foucault, and follows the work of science studies scholars, beginning with Ludwig Fleck and later Thomas Kuhn, who noted that the concept of a “discovery” of solutions or answers is a misnomer in well-designed scientific research, since outcomes of interventions or trials are already in some sense built into, embedded in the design of the research that prioritizes only this intervention.

9. RCT research for clinical trials is defined by the National Institutes of Health as being appropriate for clinical drug trials as well as epidemiological and/or behavioral interventions. However, most RCT-designed studies in epidemiological behavioral health programs (seen in public health) use some form of one of these, and often use the final (nondefined phase III model). These are:

Phase I clinical trials test a new biomedical intervention in a small group of people (e.g., 20–80) for the first time to evaluate safety (e.g., to determine a safe dosage range and to identify side effects).

Phase II clinical trials study the biomedical or behavioral intervention in a larger group of people (several hundred) to determine efficacy and to further evaluate safety.

Phase III studies investigate the efficacy of the biomedical or behavioral intervention in large groups of human subjects (from several hundred to several thousand) by comparing the intervention to other standard or experimental interventions, monitor adverse effects, and collect information that will allow the intervention to be used safely.

Phase IV studies are conducted after the intervention has been marketed. These studies are designed to monitor effectiveness of the approved intervention in the general population and to collect information about any adverse effects associated with widespread use.

NIH-Defined Phase III Clinical Trial. An NIH-defined Phase III clinical trial is a broadly based prospective Phase III clinical investigation, usually involving several hundred or more human subjects, for the purpose of evaluating an experimental intervention in comparison with a standard or controlled intervention or comparing two or more existing treatments. Often the aim of such investigation is to provide evidence leading to a scientific basis for consideration of a change in health policy or standard of care. The definition includes pharmacologic, nonpharmacologic, and behavioral interventions given for disease prevention, prophylaxis, diagnosis, or therapy. Community trials and other population-based intervention trials are also included (from NIH Public Health Service Grant <http://grants.nih.gov/grants/funding/phs398/phs398.doc>).

I also note, from talking with researchers in behavioral health research, that the notion of what constitutes phase I, II, or III research design is somewhat variable. For some, phase I entails doing the intervention to see whether there is a measurable (statistically significant) impact, not necessarily with a control group; phase II entails doing one intervention and comparing it to something else that involves a larger or different kind of intervention (likened to evaluating different “doses” of a drug); and phase III entails comparing interventions and using a control (or placebo or “no”) intervention.

10. Although it is new, I believe the *Global Public Health* journal (Columbia University) may prove an exception to the norm.

11. I refer here to the *International Journal of Public Health*, the *Journal of Health Services*, the *WHO Bulletin*, even *Social Science and Medicine*, and other international health bulletins.

12. Many of my examples come from the research proposals I have reviewed for various UCSF and NIH funding agencies and for projects within UCSF’s Global Health Sciences.

13. It is, in part, as a result of the social scientific critique of failed health development programs (Mark Nichter, Judith Justice, Arturo Escobar offer popular examples) that international health programs have adopted more rigorous standards for evaluations of interventions. These usually include not only health outcomes but cost-effectiveness outcomes.

14. James Pfeiffer explained to me (personal communication) that this model was developed by Chris Murray, who is now part of Health Alliance International, a public-private partnership in Seattle, where the focus has been to develop what they are calling “Implementation Sciences.”

15. This standardization is also produced by the critique that psychological sciences are too “soft” (as Allan Young noted in his study of PTSD).

16. The instruments include ASEBA (the Achenbach System of Empirically Based Assessment), YRBS (the Youth Risk Behavior Survey), CCSC (the Children’s Coping Strategies Checklist), SSIS (adapted from Social Skills Rating System), and HSSSC (the Harter Social Support Scale for Children).

17. Cited in Adams 2005.

18. She is referring to Thomas Kuhn's *The Structure of Scientific Revolutions* (University of Chicago Press, 1996 [1962]), and Ian Hacking's *The Social Construction of What?* (Harvard University Press, 2000). This quotation is taken from Kelly Knight's doctoral dissertation, "Tricky: Evidence, Facts and Reproduction among Homeless Women Addicts in California" (University of California San Francisco, Program in Medical Anthropology 2010).

3

The “Right to Know” or “Know Your Rights”?

Human Rights and a People-Centered Approach to Health Policy

JOSEPH J. AMON

Human rights abuses fuel vulnerability to HIV infection and act as barriers to universal access to prevention, treatment, and care. This has been recognized in numerous international declarations, and attention to human rights has been incorporated into the mission statements and work plans of grassroots groups and global organizations alike. Yet *how* recognition of this relationship is translated into action varies, as do the ways in which the *language* of human rights has been employed by HIV programs and campaigns.

“Treatment” activists, human rights advocates, and HIV programs of varying types (from evangelical and other faith-based organizations to youth groups to multinational development organizations) sometimes unite to promote rights (such as the right to treatment) and sometimes diverge in their interpretation of rights (e.g., the right to condoms or methadone as an effective HIV prevention strategy, or the rights of LGBT people). United Nations agencies and governments, including those governments traditionally less sympathetic to human rights, have increasingly been forced to define and defend their HIV efforts using the language, if not the principles, of human rights.